



Humane paper review

Generative Agents: Interactive Simulacra of Human Behavior

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Generative Agents: Interactive Simulacra of Human Behavior

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UIST 2023

2026.07.03

Research Gap

- Believable human behavior: a long-standing goal
 - Game NPCs, social prototyping, interactive worlds
- Prior approaches fail in open-ended worlds
 - Rule-based (FSM, behavior trees): hand-authored, narrow
 - Cognitive architectures / RL: no open-ended social behavior

Research Gap

- LLMs know human behavior, but...
 - Limited context window → cannot hold days of experience
 - No persistent memory → long-term coherence breaks down

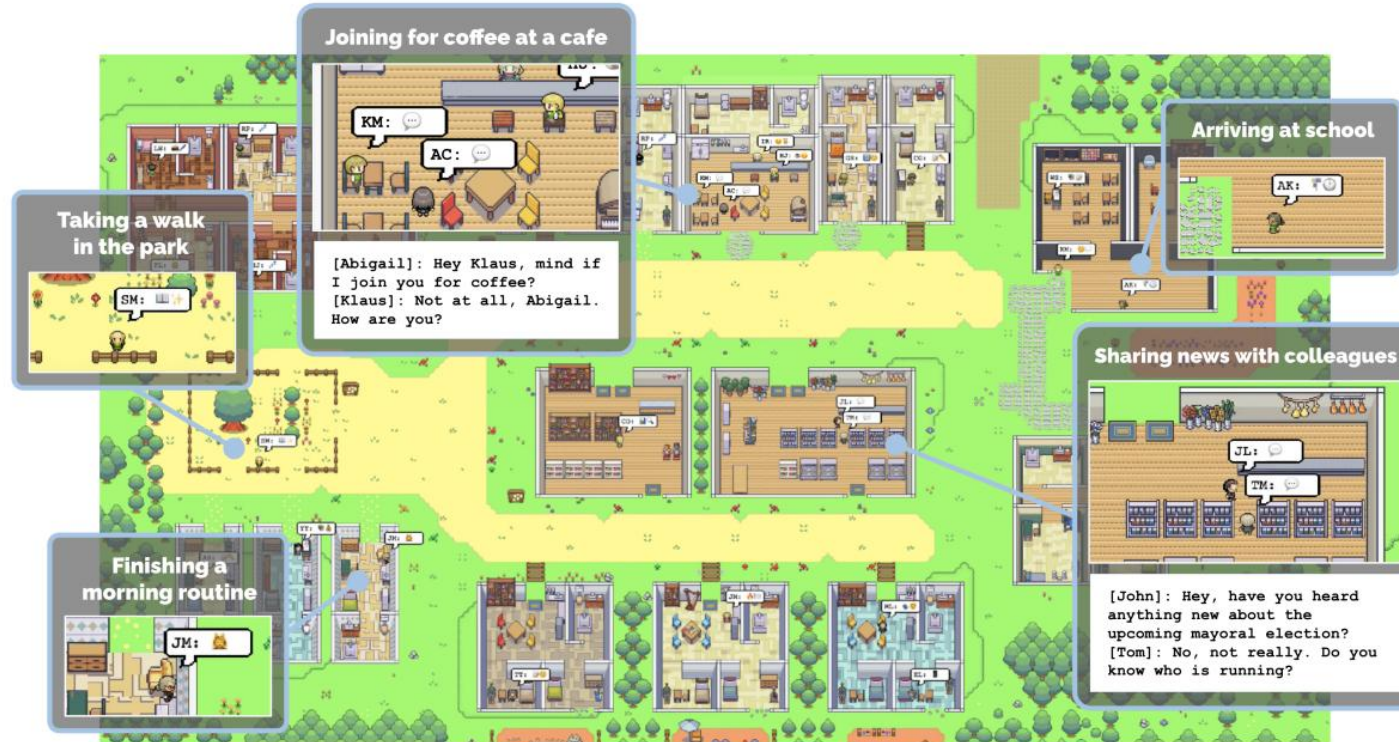
This work: an **agent architecture around an LLM** that remembers, reflects, and plans

Contributions

1. Generative agent architecture
 - Memory stream + Retrieval + Reflection + Planning
2. Smallville: interactive sandbox demo
 - 25 agents, 2 game-days → emergent social behavior
3. Two evaluations
 - Controlled ablation (interviews) + end-to-end analysis

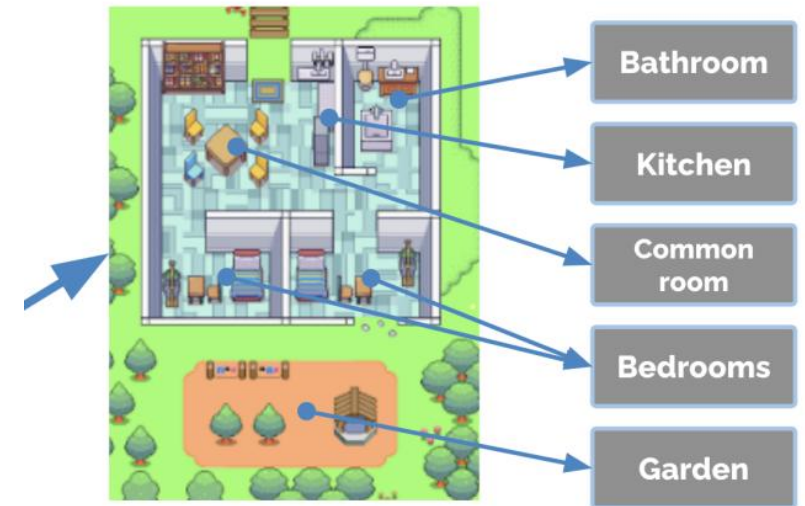
Smallville: The Sandbox World

- A 2D village: cafe, bar, shops, school, dorm, park...
- 25 agents, each seeded with one paragraph of identity

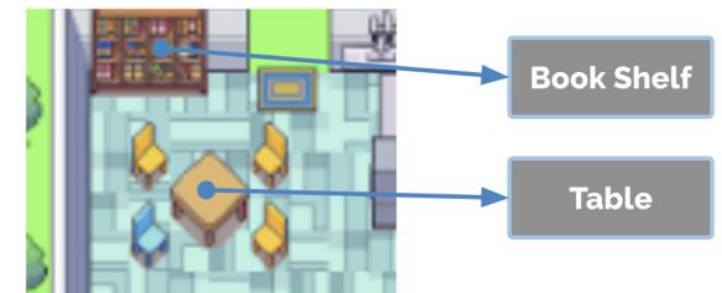


The World Is a Tree

- Town → areas → rooms → objects
- Rendered into natural language for prompts
- Each agent keeps only its own subtree
 - Grows with exploration — and can become outdated



Family House



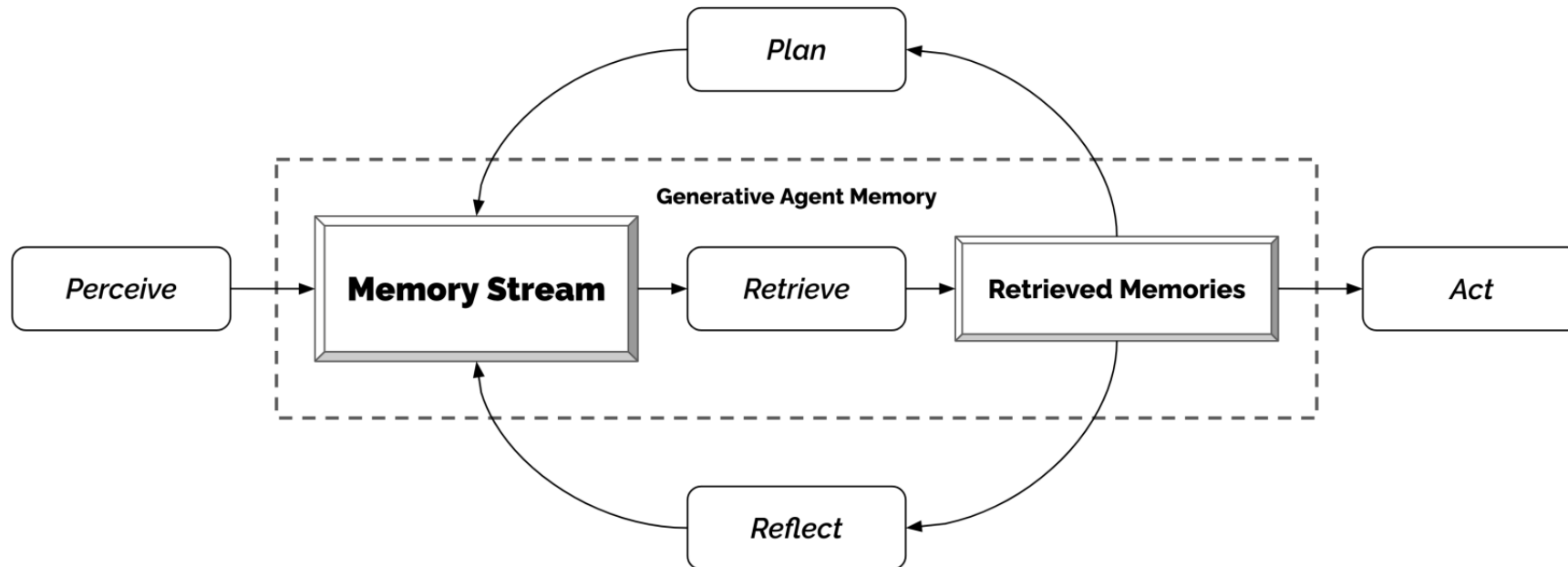
Common Room

Actions & User Interventions

- Actions are natural language sentences
 - → emoji display / location mapping / pathfinding
- Users can intervene in three ways
 - Chat as a persona
 - Speak as the agent's "inner voice"
 - Edit object states (stove → "burning" → John puts it out)

Generative Agent Architecture

- Perceive → Memory Stream → Retrieve → Act
- One LLM, many prompted calls — the agent is a pipeline



Code View: Generative Agent Architecture

```
# persona.py / Persona.move()  
  
perceived = self.perceive(maze)  
retrieved = self.retrieve(perceived)  
plan = self.plan(maze, personas, new_day, retrieved)  
self.reflect()  
  
return self.execute(maze, personas, plan)
```

Component 1: Memory Stream

- Every experience stored as a natural language record
- Memory = (description, created time, last-accessed time)
- Three types in one stream:
 - Observation · Reflection · Plan

Code View: Memory Stream

```
# associative_memory.py / ConceptNode

self.type = node_type      # event / thought / chat
self.created = created
self.last_accessed = self.created

self.subject = s
self.predicate = p
self.object = o

self.description = description
self.embedding_key = embedding_key
self.poignancy = poignancy
self.keywords = keywords
self.filling = filling
```

Retrieval: Which Memories Matter Now?

$$\textit{score} = \textit{recency} + \textit{importance} + \textit{relevance}$$

- Recency — exponential decay (0.995) since last access
- Importance — LLM rates 1–10 (brushing teeth = 1, breakup = 10)
- Relevance — embedding similarity to the current context
- Top-scored memories fill the LLM context window

Code View: Retrieval Score

```
# retrieve.py / new_retrieve()

recency_out = extract_recency(persona, nodes)
importance_out = extract_importance(persona, nodes)
relevance_out = extract_relevance(persona, nodes, focal_pt)

gw = [0.5, 3, 2]

master_out[key] = (
    persona.scratch.recency_w * recency_out[key] * gw[0]
+ persona.scratch.relevance_w * relevance_out[key] * gw[1]
+ persona.scratch.importance_w * importance_out[key] * gw[2]
)

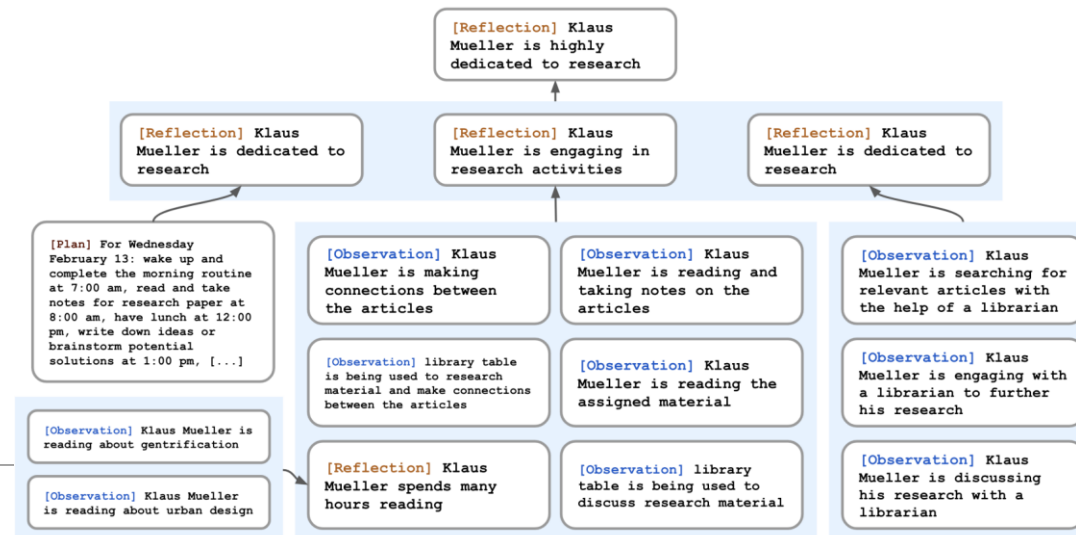
master_nodes = top_highest_x_values(master_out, n_count)
```

Component 2: Reflection — Why?

- Raw observations only support shallow inference
- Agents need higher-level, abstract conclusions
- Trigger: Σ importance of recent events > 150
 - 2–3 times per day in practice

Reflection: Two Steps

- Step 1 — Ask: from the 100 most recent memories, generate 3 salient high-level questions
- Step 2 — Answer: retrieve per question → extract 5 insights with evidence pointers
- Reflections are stored as memories → a tree of abstraction



Code View: Reflection

```
# reflect.py / run_reflect()

focal_points = generate_focal_points(persona, 3)
retrieved = new_retrieve(persona, focal_points)

for focal_pt, nodes in retrieved.items():
    thoughts = generate_insights_and_evidence(persona, nodes, 5)

    for thought, evidence in thoughts.items():
        thought_poignancy = generate_poig_score(
            persona, "thought", thought
        )
        persona.a_mem.add_thought(
            created, expiration,
            s, p, o,
            thought,
            keywords,
            thought_poignancy,
            (thought, get_embedding(thought)),
            evidence
        )
```

Reflection in Action: Klaus & Maria

"Who would you want to spend time with?"

With reflection

Klaus names Maria —
their shared research passion

Without reflection

Names whoever he
bumped into most often

Component 3: Planning — Why?

- **Moment-by-moment generation breaks coherence**
 - e.g., lunch at 12:00 ... and lunch again at 12:30
- Plans commit the agent to a coherent future
- Plans are stored in the memory stream too

Code View: Planning

```
# plan.py / plan()

focused_event = _choose_retrieved(persona, retrieved)
reaction_mode = _should_react(persona, focused_event, personas)

if reaction_mode[:9] == "chat with":
    _chat_react(maze, persona, focused_event,
               reaction_mode, personas)

elif reaction_mode[:4] == "wait":
    _wait_react(persona, reaction_mode)
```

```
# plan.py / _chat_react()

convo, duration_min = generate_convo(
    maze, init_persona, target_persona
)
convo_summary = generate_convo_summary(init_persona, convo)

_create_react(init_persona, ...)
_create_react(target_persona, ...)
```

Planning: Recursive Decomposition

Day outline (5–8 chunks)



Hour-level chunks



5–15 minute actions

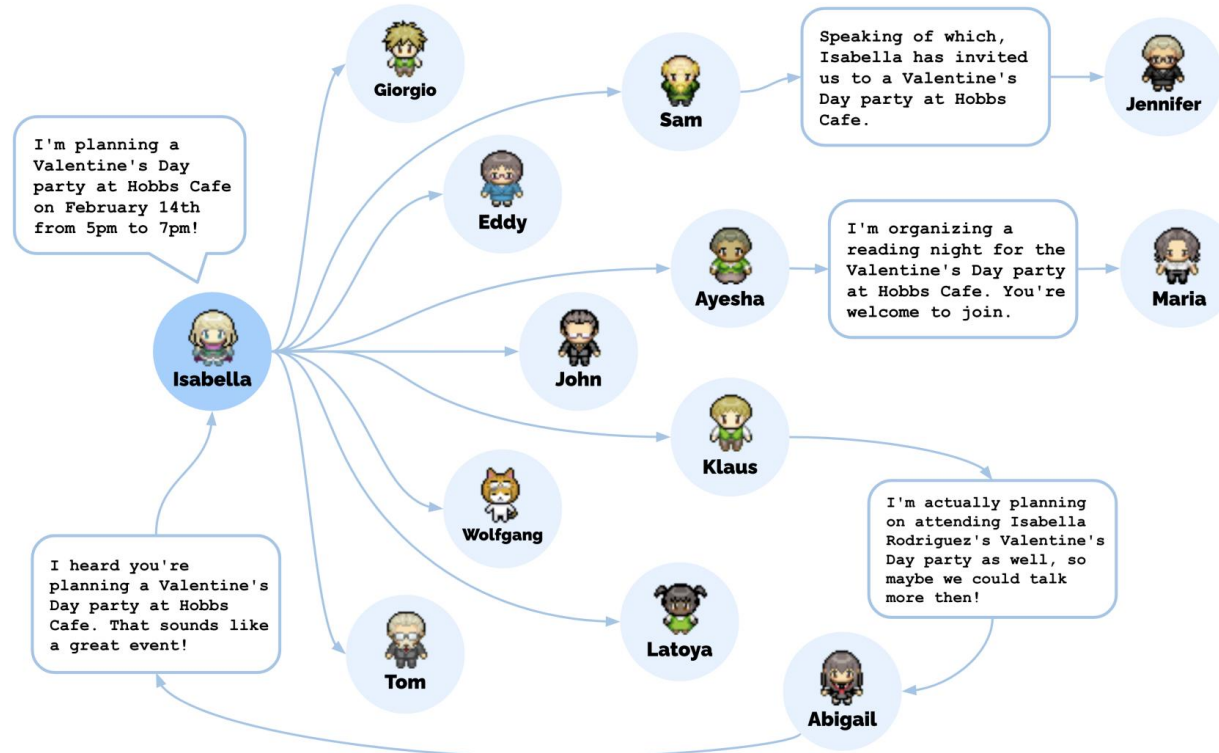
- Generated from the agent's cached "summary description" + yesterday's summary
- Summary description reused across prompts (cost control)

Reacting & Dialogue

- Every timestep: observe → "should I react?" → re-plan from now
 - John hears about Eddy's composition → inserts a chat, shifts schedule
- Dialogue: each turn conditioned on retrieved memories of the partner
 - Knowledge asymmetry between speakers emerges naturally

Emergent Behavior: The Valentine's Party

- Only one seed: Isabella "wants to throw a party"
- Invitations spread · Maria invites her crush Klaus · decorations planned
- 5 agents actually show up at the cafe, Feb 14, 5 pm



Emergence in Numbers (2 Game-Days)

4% → 32%

mayor candidacy awareness

4% → 52%

party awareness

0.167 → 0.74

network density

*Of 12 who heard about the party: 5 attended, 3 had conflicts,
4 "considered but never planned"*

Evaluation 1: Interview the Agents

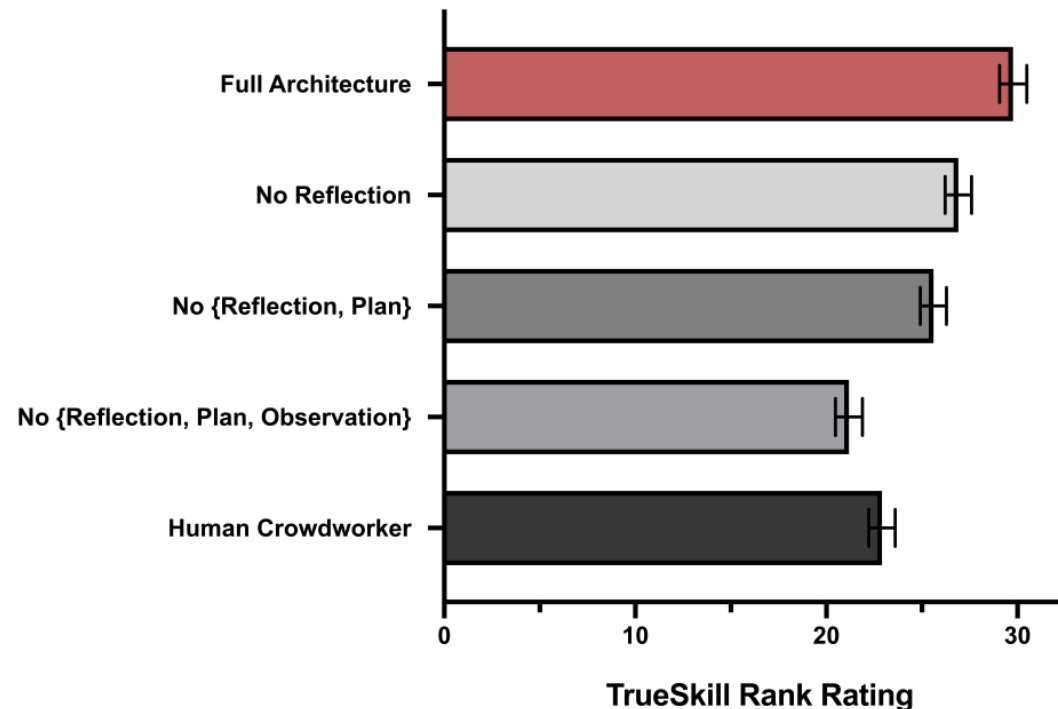
- After 2 game-days, each agent answers 25 questions
- 5 areas: Self-knowledge / Memory / Plans / Reactions / Reflection
- 100 human evaluators rank believability → TrueSkill ratings
- Kruskal-Wallis + Dunn post-hoc tests

Ablation Conditions

Condition	Memory access
Full architecture	Memory + Reflection + Planning
No reflection	Observation + Planning
No reflection, no planning	Observation only
No memory	Interview context only
Human crowdworker	Watches replay, answers in character (baseline)

Evaluation 1: Results

- Full > no-reflection > no-planning > no-memory $\approx$ human
- Monotonic drop $\rightarrow$ every component contributes
- Full agents rated more believable than human-authored answers



Observed Error Patterns

- Retrieval failure
 - Knew it, but couldn't recall it
- Embellishment
 - Adds details that were never in memory
- World-knowledge leakage
 - Neighbor "Adam Smith" described as... the economist

Evaluation 2: End-to-End

- Full 25-agent, 2-day simulation → measure emergence
- Hallucination check: "yes" answers without an actual memory record were excluded
- Relationship answers: only ~1.3% hallucinated

Diffusion · relationships · coordination — all emerged, all verified

Limitations (1/2)

- Location choice degrades as knowledge grows
 - Larger subtree → noisier choices (bar treated as a generic cafe)
- Physical norms missing from natural language
 - Agents walk into a one-person dorm bathroom; enter closed shops

Limitations (2/2)

- Instruction-tuning artifacts
 - Overly polite & cooperative; tone never deepens with relationships
- Cost & scale
 - Thousands of USD, days of real time — for 25 agents × 2 days
 - Long-run stability untested

Discussion: Applications & Ethics

Applications

- Social simulation & what-ifs
- Design prototyping
- Game NPCs

Ethical risks

- Parasocial bonds → require disclosure
- Believable ≠ accurate human modeling
- Impersonation → audit logs
- Complement, not replace, real humans

Takeaway

- 1) Memory + Reflection + Planning → long-term believable behavior
- 2) Every component contributes (monotonic ablation)
- 3) Social behavior emerges from individual agents

The memory–reflection–planning pattern became the **template for LLM-agent research**

Review

Strengths

- Architecture-level, model-agnostic contribution
- Emergent social dynamics demonstrated
- Honest failure analysis; hallucination-checked metrics

Weaknesses

- Single run, 25 agents, 2 days
- Believability is subjective, interview-based
- Hallucination not prevented in-system; high cost